Let $\eta_1 = pF - \mu_1\xi_1^2\xi_2/2$, $\eta_2 = p^2 + \mu_1\xi_1^2/2$, so $\lambda = \eta_1/\eta_2$, then calculate

$$\frac{\mathrm{d}}{\mathrm{d}p}\lambda = (\eta_2 F - 2\eta_1 p)/\eta_2^2,$$

$$\frac{\mathrm{d}}{\mathrm{d}\xi_1}\lambda = (-\mu_1\xi_1\xi_2\eta_2 - \mu_1\xi_1\eta_1)/\eta_2^2,$$

to see that

$$\begin{aligned} p\frac{\mathrm{d}\lambda}{\mathrm{d}p} + \xi_1\frac{\mathrm{d}\lambda}{\mathrm{d}\xi_1} &= \frac{(pF - \mu_1\xi_1^2\xi_2)\eta_2 - (2p^2 + \mu_1\xi_1^2)\eta_1}{\eta_2^2} \\ &= 2\left[-\frac{pF}{2\eta_2} + \lambda - \lambda\right] = -\frac{pF}{\eta_2}, \end{aligned}$$

hence

$$\begin{aligned} \mathrm{div}(\boldsymbol{f}) &= \frac{pF}{\eta_2} - 2\lambda - \xi_2 \\ &= \frac{pF - 2(pF - \mu_1\xi_1^2\xi_2/2)}{p^2 + \mu_1\xi_1^2/2} - \xi_2 \\ &= \frac{-pF + \mu_1\xi_1^2\xi_2}{p^2 + \mu_1\xi_1^2/2} - \xi_2. \end{aligned}$$

We find that a solution of the equation $\nabla w \cdot \boldsymbol{f} = \mathrm{div}(\boldsymbol{f})$ is then given by

$$w = \frac{U + \frac{1}{2}\xi_2^2}{p^2 + \frac{1}{2}\mu_1\xi_1^2}.$$

Integrating over the isokinetic surface $p^2 + \frac{1}{2}\mu_1\xi_1^2 = \beta^{-1}$, and with respect to the variable ξ_2, yields the Gibbs configurational density.

The exciting prospect here is that we now have a new variable ξ_2 which has, in equilibrium, a Gaussian distribution. The Ornstein-Uhlenbeck terms are therefore compatible with this extended distribution and can be incorporated (as in Nosé-Hoover-Langevin) to provide ergodicity and control of the convergence to equilibrium.